Class Exercise 11: Data Analysis & Graphing

Instructions: **Report** the statistical results of our study and **Graph** your results

Requirements: To successfully complete this exercise, you are required to:

1. **Identify and report** statical information from your Jamovi analysis output
2. **create** visual representations of the data result pattern(s)

Please use a **word processing program** (such as MS Word) to complete this exercise.

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Preliminary Results - analyzed via Pearson's *r* Correlation

Let's start with testing whether there is an association between IH and apology quality. Fill out the following information and then describe the statistical results. This data will be used for your Paper Results' section.

1. Statistical test conducted: Pearson's *r* correlation
2. $r(104) = 0.37$, $p < .001$

**Notes: If p very small, you can just put p < .001. Otherwise, put the exact p-value (p = .xxx) and round to 3 decimals. Round r to 2 decimals*

The number in parentheses is the degrees of freedom. The number after the = sign is the correlation coefficient, and the reported p-value indicates whether the correlation coefficient is statistically significant.

Primary Results - analyzed via ANOVA

Let's finish our results prep with what we want to report in our paper. Fill out the following information and then describe the statistical results. This data will be used for your Paper Results' section. *M* refers to the Mean or Average, *SD* refers to the Standard Deviation, and *SE* refers to the Standard Error (you will only need the *SE* for graphing purposes)

1. IV 1: Intellectual Humility (IH vs. Control)
2. IV 2: Post-Transgression Response (Unforgiving vs. Forgiving)
3. DV: Apology Quality
4. Statistical test performed = 2 x 2 between-subjects ANOVA
   a. Main effect of Intellectual Humility
      i. IH
         1. $M = 5.08$

            2. *SD* = 0.94
            3. *SE = 0.15*
        ii. Control
            1. *M* = 4.93
            2. *SD* = 1.11
            3. *SE = 0.14*
        iii. Inferential Stats: *F*(1, 103) = 0.52, *p* = .473
    b. Post-Transgression Response
        i. Unforgiving
            1. *M* = 5.04
            2. *SD* = 1.18
        ii. Forgiving
            1. *M* = 4.95
            2. *SD* = 0.92
        iii. Inferential Stats: *F*(1, 103) = 0.22, *p* = .644
    c. Interaction effect
        i. Unforgiving
            1. IH: *M* =5.16 , *SD* = 0.89
                a. *SE* = 0.20
            2. Control: *M* =4.96 , *SD* = 1.36
                a. *SE* = 0.25
        ii. Forgiving
            1. IH: *M* =5.01 , *SD* = 1.01
                a. *SE* = 0.22
            2. Control: *M* = 4.91, *SD* = 0.87
                a. *SE* = 0.15
        iii. Inferential Stats: *F*(1, 103) = 0.06, *p* = .815

***We will hold onto this statistical information and work to put this information in paragraph format (i.e. results paragraphs) in the next CEs**

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Below is example output from different data, but there's additional information about the other statistics in our output if you want to take a deeper dive.

ANOVA - Apology likelihood

	Sum of Squares	df	Mean Square	F	p	$\eta^2 p$
severity_recode	28.599	1	28.599	14.675	< .001	0.080
empathy_recode	14.573	1	14.573	7.478	0.007	0.042
severity_recode ✻ empathy_recode	0.087	1	0.087	0.045	0.833	0.000
Residuals	329.352	169	1.949			

[3]

Post Hoc Tests

Post Hoc Comparisons - severity_recode ✻ empathy_recode

Comparison											95% Confidence Interval	
severity_recode	empathy_recode		severity_recode	empathy_recode	Mean Difference	SE	df	t	p_{tukey}	Cohen's d	Lower	Upper
Low severity	High empathy	-	Low severity	Low empathy	0.546	0.279	169.000	1.953	0.210	0.391	−0.007	0.788
		-	High severity	High empathy	−0.874	0.305	169.000	−2.863	0.024	−0.626	−1.063	−0.189
		-	High severity	Low empathy	−0.237	0.320	169.000	−0.741	0.880	−0.170	−0.623	0.283
	Low empathy	-	High severity	High empathy	−1.420	0.291	169.000	−4.877	< .001	1.017	0.591	1.443
		-	High severity	Low empathy	−0.783	0.306	169.000	−2.555	0.055	−0.561	−0.998	−0.123
High severity	High empathy	-	High severity	Low empathy	0.637	0.330	169.000	1.930	0.219	0.456	−0.013	0.926

Note. Comparisons are based on estimated marginal means

Two-way ANOVA, Post Hoc Tests

- For the main effects of empathy and severity, if the *F*-test is significant, then we can look at our descriptive statistics to determine which condition had a higher mean than the other.
- For the **interaction**, we can look at the post hoc tests to determine which conditions differed from each other.
  - **We would only interpret these results if the interaction was statistically significant.** However, the interaction between empathy and severity on apology likelihood was not significant, $F(1, 169) = .05$, $p = .833$, $\eta_p^2 < .001$.
  - Look at the *p*-values, or *ptukey* in this case, which adjusts for multiple comparisons to avoid inflating our type 1 error rate.
  - Pretending for a moment that the interaction was significant (which it's not), look at the comparisons that are relevant to your hypotheses. Here, it was hypothesized that empathy's effect on apology likelihood would be stronger for high-severity offenses than for low-severity offenses. So, we would look at the difference between **high and low empathy** within each **severity level**.

Results of the two-way ANOVA

ANOVA Table

- **F** is the *F*-value.
  - It's the ratio of between-group variance (differences among the group means) to within-group variance (variability within each group). A larger *F*-value indicates the differences between group means are large relative to the variation within groups, which increases the likelihood that the test will produce a statistically significant result (if $p < .05$).
  - There are two values for the degrees of freedom (df) associated with the F-value for each effect: the numerator df, which is 1, and the residual or denominator df, which is 180.
- ***p* (or the p-value)** tells us whether the mean difference is statistically significant!
  - Here, our *p*-values are < .001, .007, and .833
  - If this value is **< .05,** then there's a statistically significant difference among conditions.
  - Our *p*-value tells us whether there **is** a difference, but <u>NOT</u> **how** groups are different**.** To understand how the groups differ, we need to look at the descriptive stats for the main effects and the post-hoc tests for the interaction.
- η_p^2–or partial eta squared—is an effect size. Partial eta-squared is the proportion of variance explained by an IV after accounting for the variance explained by other IVs in the ANOVA.

Graphing:

Paste your 3 graphs here: Interaction, main effect of IH, and main effect of post-transgression response

APA Checklist & Tips!

- Use colors that are easy to read.
- Ensure all text in the graph is in **Times New Roman, 12pt font**.
- Add **x-axis and y-axis lines**, ensuring the **y-axis reflects the scale range** for the dependent variable (DV).
- **Add labels** to the x-axis and y-axis.
- Include **data labels** on the outside end of each bar to display the means.
- Make all **text and lines black** for consistency.
- The background lines may be removed, but it's okay to keep them if necessary.
- Include a **note** at the bottom stating that the error bars represent standard errors.
- Ensure the **figure # and title** are in APA format (see example below)